



HOSPITAL EMERGENCY ROOM DASHBOARD

End-to-End Power BI Dashboard Project



Transforming Emergency Room Data
into **Actionable Business Insights**

INTERACTIVE DASHBOARD PREVIEW



Created by
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Power BI



Power Query



DAX



Data Modeling



Data
Visualization



BUSINESS PROBLEM



Emergency departments generate thousands of patient records every month, making it challenging to monitor operational performance and identify areas for improvement.

KEY QUESTIONS

- ✓ How many patients visited the Emergency Room?
- ✓ Is the average waiting time meeting the 30-minute target?
- ✓ Which departments receive the highest patient referrals?
- ✓ Which age groups and demographics require the most care?
- ✓ Which days and hours experience the highest patient volume?
- ✓ How satisfied are patients with the service provided?



SOLUTION

To address these challenges, I developed an interactive **Power BI dashboard** that transforms raw hospital data into meaningful business insights, enabling management to monitor KPIs, analyze trends, identify operational bottlenecks, and support **data-driven decision-making**.



Power BI

WHAT THE DASHBOARD DELIVERS



Monitor
monthly and long-term performance



Analyze
patient waiting times



Track
admissions and referrals



Understand
patient demographics



Identify
peak hours and busiest days



Explore
patient-level records for deeper analysis

“ Better Insights. Smarter Decisions. Better Patient Care. ”



CONSOLIDATED VIEW

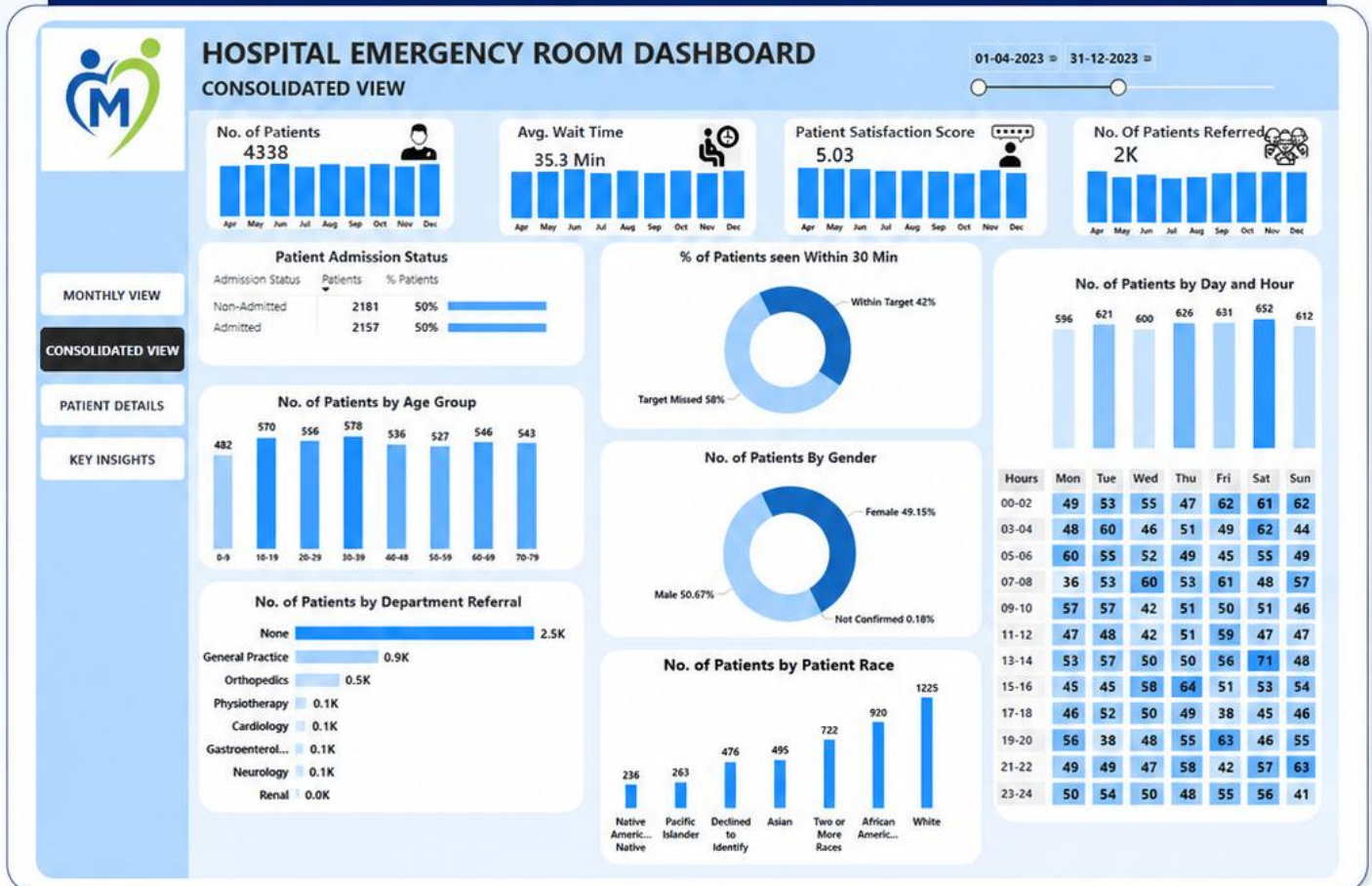
Analyze Any Date Range. Uncover **Powerful Trends**.



Any Date Range

01-04-2023 to
31-12-2023

The Consolidated View provides a complete overview of ER performance across **any selected period**, helping management make data-driven decisions.



Key Highlights

- ✓ Total 4,338 patients in the selected period
- ✓ Average wait time is 35.3 minutes
- ✓ 58% of patients exceeded the 30-minute target
- ✓ Admission status is evenly split (50% / 50%)
- ✓ Weekend hours show higher patient volume
- ✓ Detailed breakdown by age, gender, race & department

Why It Matters

- Understand long-term trends across any time period
- Identify peak hours and busy days
- Improve resource allocation and operational efficiency
- Make faster, data-driven decisions



This view helps hospital management **track performance KPIs**, **spot bottlenecks**, and **improve patient experience**.



MONTHLY PERFORMANCE DASHBOARD



Complete Monthly Overview of Emergency Room Performance



Total Patients

469

▲ Overview of all patients admitted



Average Wait Time

35.3 Min

▼ Monitor wait time performance



Patient Satisfaction Score

4.63 /10

▲ Measure patient experience



Patients Referred

185

▲ Track department referrals



HOSPITAL EMERGENCY ROOM DASHBOARD

MONTHLY VIEW

Apr 2024

Last Refresh : 15-07-2026 07:31:12

Year

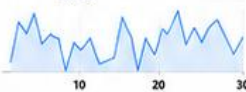
2024

Month Name

Apr

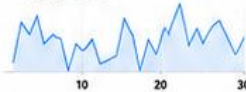
No. of Patients

469



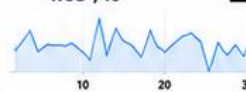
Avg. Wait Time

35.3 Min



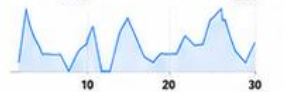
Patient Satisfaction Score

4.63 /10



No. Of Patients Referred

185



MONTHLY VIEW

CONSOLIDATED VIEW

PATIENT DETAILS

KEY INSIGHTS

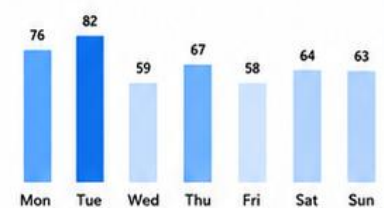
Patient Admission Status

Admission Status	Patients	% Patients
Non-Admitted	252	54%
Admitted	217	46%

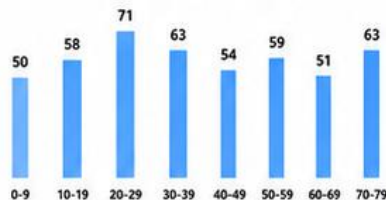
% of Patients seen Within 30 Min



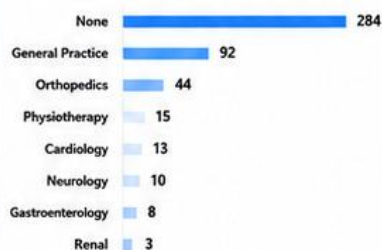
No. of Patients by Day and Hour



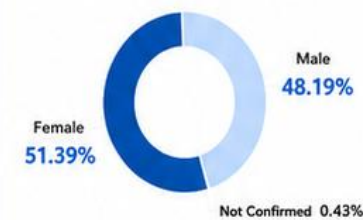
No. of Patients by Age Group



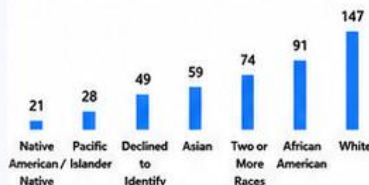
No. of Patients by Department Referral



No. of Patients By Gender



No. of Patients by Patient Race



Hours	Mon	Tue	Wed	Thu	Fri	Sat	Sun
00-02	3	10	2	6	1	5	5
03-04	6	4	4	4	8	7	3
05-06	11	7	7	10	2	4	7
07-08	3	9	5	9	3	3	3
09-10	5	6	6	6	6	11	4
11-12	7	6	4	3	7	4	4
13-14	5	8	5	3	4	5	6
15-16	8	7	7	7	3	4	5
17-18	6	6	5	3	6	5	7
19-20	8	4	4	2	6	6	3
21-22	7	9	3	7	6	3	6
23-24	7	6	7	7	6	7	10

BUSINESS HIGHLIGHTS



Monitor Patient Admissions

Track admitted vs. non-admitted patients in real-time.



Analyze Waiting Time Performance

Evaluate average wait time against the 30-minute target.



Identify Referral Patterns

Understand which departments receive the most referrals.



Track Operational Efficiency

Measure satisfaction, patient flow, and peak hours to improve efficiency.



This dashboard empowers hospital management with real-time insights to **improve patient experience** and **operational decisions**.



Power BI

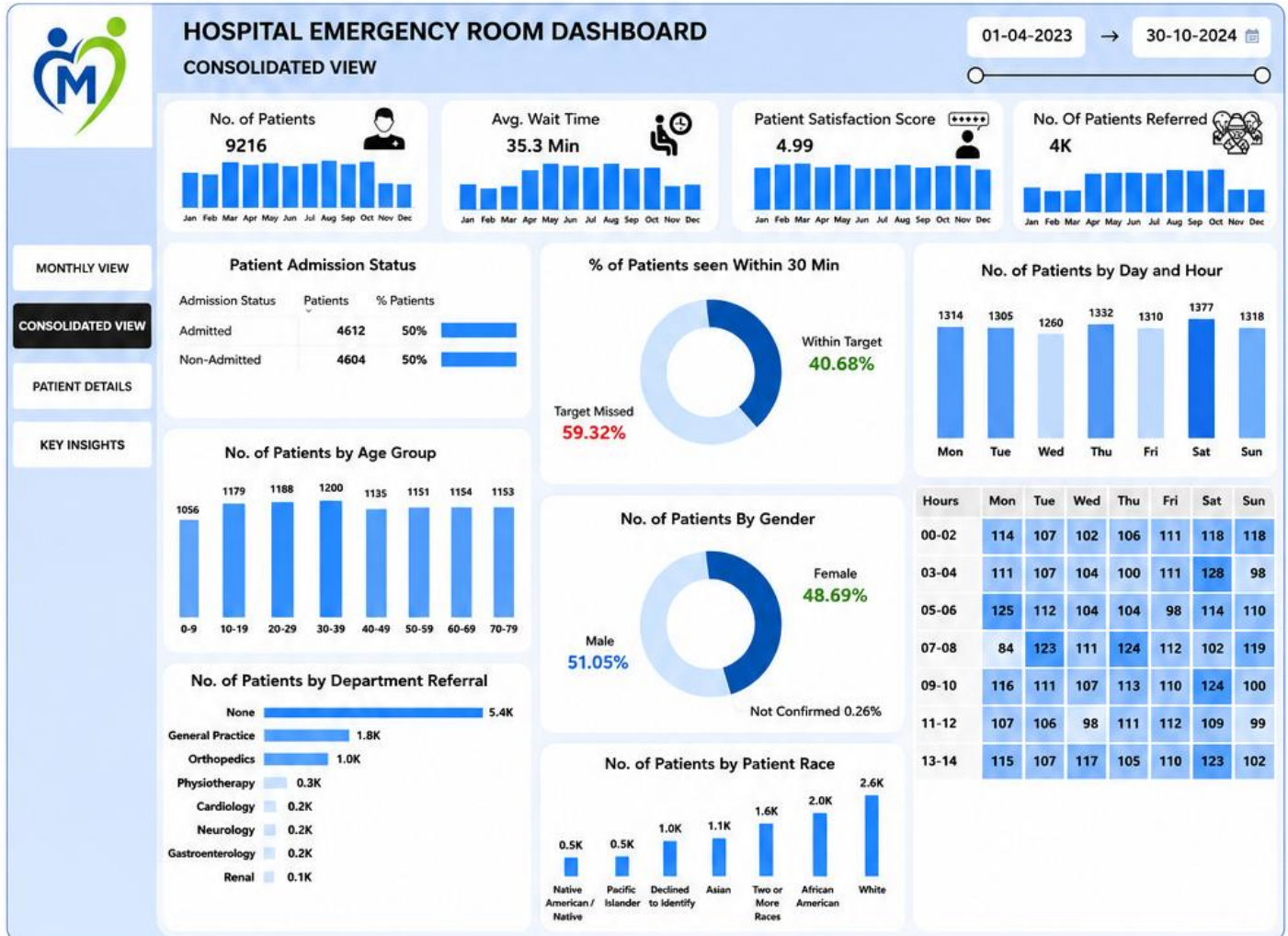


KEY BUSINESS INSIGHTS

Turning Hospital Data into Actionable Decisions



★ These insights help improve patient experience, optimize resources, and support data-driven decision-making.



WAIT TIME

- Average wait time is **35.3 minutes**, exceeding the **30-minute** operational target.
- Only **40.68%** of patients were seen within the target, while **59.32%** missed the target – our biggest operational concern.



PEAK LOAD

- Saturday (1,322)** and **Tuesday (1,318)** experience the highest patient volume.
- Peak admission hours occur around **11 AM** and **7 PM** – two major demand windows for staffing.



REFERRALS

- General Practice (1.8K)** receives the highest number of referrals.
- Followed by **Orthopedics (1.0K)**. This helps prioritize specialist availability and resource allocation.



DEMOGRAPHICS

- Adults aged **30–39 years (1,200)** represent the largest patient group.
- Gender distribution is balanced:

Male **51.05%**

Female **48.69%**

Not Confirmed **0.26%**



ADMISSIONS

- Total **9,216** patients analyzed over the selected period.
- Admission status is almost evenly distributed:



ADMITTED

4,612

(50%)



NON-ADMITTED

4,604

(50%)

This indicates a balanced mix of critical and non-critical emergency cases.



KEY TAKEAWAY

Waiting time is the **primary operational challenge**, while **peak load**, **referral patterns**, and **demographics** provide clear opportunities to **improve efficiency and patient care**.



Data-driven insights. Better decisions.
Better patient outcomes.

Built with



Power BI



RATING

9.8 / 10





PATIENT DETAILS & TECHNICAL IMPLEMENTATION



From High-Level Insights to Patient-Level Analysis

Decision-makers often need to investigate individual patient records beyond summary dashboards. This page provides detailed patient-level information with interactive filtering, enabling quick analysis and informed decision-making.



SELECT DATE RANGE

28-08-2023 → 21-02-2024

Choose any start and end date to view patient details



HOSPITAL EMERGENCY ROOM DASHBOARD PATIENT DETAILS

28-08-2023 → 21-02-2024

MONTHLY VIEW

CONSOLIDATED VIEW

PATIENT DETAILS

KEY INSIGHTS

Patient Id	Patient Name	Patient Gender	Patient Age	Year	Quarter	Month	Day	Patient Race	Patient Waittime	Admission Status	Department Referral
100-17-5081	V Flicker	Male	67	2024	Qtr 1	January	14	African American	60	Non-Admitted	None
100-70-0071	R Downham	Male	38	2024	Qtr 1	January	14	African American	57	Non-Admitted	None
103-04-9687	E Finby	Male	48	2023	Qtr 4	December	23	African American	29	Non-Admitted	None
103-34-9671	S Brissard	Male	38	2023	Qtr 3	September	24	African American	54	Admitted	None
103-73-6870	Q Pickston	Male	15	2023	Qtr 4	October	24	African American	22	Non-Admitted	None
103-87-1157	Y Pashby	Female	51	2023	Qtr 3	September	15	African American	46	Admitted	None
106-76-7225	A Gipson	Female	17	2023	Qtr 3	September	14	African American	44	Admitted	None
107-69-8839	O Plane	Male	32	2023	Qtr 4	December	26	African American	51	Admitted	None
110-34-1005	V Butting	Not Confirmed	5	2023	Qtr 4	December	28	African American	31	Non-Admitted	Orthopedics
110-77-9670	D Skacel	Male	36	2023	Qtr 4	December	10	African American	42	Admitted	Orthopedics
111-90-3138	P Claige	Male	1	2023	Qtr 4	October	17	African American	10	Admitted	Orthopedics
112-04-3574	X Whyley	Female	56	2023	Qtr 4	October	17	African American	25	Admitted	None
113-95-3971	B Preto	Male	1	2023	Qtr 4	October	28	African American	25	Non-Admitted	Orthopedics
114-16-7000	J Kimm	Male	27	2023	Qtr 4	November	7	African American	45	Non-Admitted	General Practice
114-58-3401	L Meiningen	Male	65	2023	Qtr 4	November	6	African American	28	Admitted	Physiotherapy
115-06-8540	W Jouannisson	Male	39	2023	Qtr 3	September	19	African American	21	Non-Admitted	General Practice
115-43-2057	R Renbold	Male	11	2023	Qtr 4	December	6	African American	13	Admitted	None
120-35-8220	G Cromack	Female	69	2023	Qtr 4	November	26	African American	30	Admitted	Orthopedics
121-36-1815	Q Antoons	Male	76	2024	Qtr 1	February	7	African American	40	Admitted	None

PATIENT DETAILS INCLUDED



Patient Information

Patient ID, Name, Age & Gender



Admission Details

Admission Status, Admission Date, Wait Time



Medical Information

Department Referral, Patient Race, Visit Timeline



Interactive Filtering

Custom date range selection, dynamic record filtering for detailed patient-level view

TECHNICAL SKILLS DEMONSTRATED



POWER BI

- ✓ Interactive Dashboard Design
- ✓ Table Visualization
- ✓ Navigation Buttons
- ✓ Slicers & Filters
- ✓ Report Layout Design



DAX

- ✓ Measures
- ✓ Calculated Columns
- ✓ CALCULATE()
- ✓ SWITCH()
- ✓ DIVIDE()
- ✓ Date Intelligence



POWER QUERY

- ✓ Data Cleaning
- ✓ Data Transformation
- ✓ Date Table Creation
- ✓ Data Type Optimization
- ✓ Last Refresh Timestamp

BUSINESS VALUE



Investigate individual patient records with complete accuracy.



Improve operational transparency and track patient journey effectively.



Support detailed reporting, compliance and internal auditing needs.



Enable faster, data-driven decisions to improve patient care and efficiency.



RATING

9.8 / 10 ★★★★★

This dashboard bridges executive dashboards and raw patient data, enabling **deep analysis** and **better operational outcomes**.



Power BI



PROJECT LEARNINGS

Turning Experience into Growth



While building this project, I strengthened my understanding of key concepts and tools in **Business Intelligence** and **Data Analytics**.



KEY LEARNINGS



BUSINESS INTELLIGENCE

Learned how to analyze healthcare data and convert it into actionable insights that support better business decisions.



DASHBOARD DESIGN PRINCIPLES

Applied design best practices to build clean, user-friendly, and interactive dashboards for effective data communication.



DAX CALCULATIONS

Created measures and calculated columns using advanced DAX functions to perform complex business logic.



DATA MODELING

Built a strong data model with proper relationships to ensure accurate and efficient analysis across multiple tables.



POWER QUERY

Used Power Query for data cleaning, transformation, and preparation to ensure data quality and consistency.



STORYTELLING WITH DATA

Learned to tell a compelling story through data by uncovering trends, patterns, and key business insights.



EXECUTIVE REPORTING

Designed executive-level reports that provide a clear overview and support strategic decision-making.



This project helped me understand how raw healthcare data can be transformed into **meaningful business insights** that drive **operational efficiency** and improve **patient outcomes**.



“

Every insight discovered is a step toward **better decisions** and a **better tomorrow**.



Power BI



RATING
9.8 / 10



PROJECT OUTCOME & BUSINESS IMPACT

From Raw Healthcare Data to Actionable Business Intelligence

This project demonstrates how data can be transformed into **meaningful insights** that support **operational efficiency**, improve **patient experience**, and enable **smarter decision-making**.



PROJECT IMPACT



1. OPERATIONAL MONITORING

- ✓ Monitor emergency room performance in real time
- ✓ Track patient volume across different periods
- ✓ Measure hospital operational efficiency



2. IMPROVE PATIENT EXPERIENCE

- ✓ Monitor average waiting time
- ✓ Identify SLA (30-minute target) performance
- ✓ Improve patient satisfaction through data-driven decisions



3. OPTIMIZE RESOURCE PLANNING

- ✓ Identify peak days and busiest hours
- ✓ Allocate doctors and staff more effectively
- ✓ Improve department referral planning



4. BETTER DECISION MAKING

- ✓ Interactive dashboards for instant analysis
- ✓ Monthly and consolidated performance comparison
- ✓ Executive-level insights for hospital management



PROJECT HIGHLIGHTS



Interactive Power BI Dashboard



Dynamic Filters & Navigation



Monthly & Date Range Analysis



Executive KPI Reporting



Advanced DAX Measures



Power Query Data Transformation



Business Storytelling with Data



KEY TAKEAWAY

“ This project showcases my ability to clean, model, analyze, and visualize real-world healthcare data using **Power BI** to deliver **actionable business insights** that support **operational** and **strategic decision-making**. ”



A data-driven approach leads to **better decisions** and **better outcomes**.

Built with



Power BI



Turning data into value for **smarter healthcare**.

TOOLS & TECHNOLOGIES USED:



Power BI



Power Query



DAX



Data Modeling



THANK YOU!

Thank you for taking the time to review my **Power BI** project.

I would truly appreciate your **feedback** and **suggestions**.



CONNECT WITH ME



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Aspiring Data Analyst



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GitHub Profile
github.com/Mohd-Momin01



Every dashboard tells a story,
every **insight** drives a decision,
and every decision creates
a **better outcome**.



If you found this project valuable,
feel free to connect and share your thoughts!

Your feedback and suggestions mean a lot and
help me improve and grow.



Let's connect, learn, and grow together
in the world of **Data Analytics!**



Power BI



SQL



Excel



Python



Data Analytics

Turning **Data** into Insights, Insights into **Impact!**